Data Preprocessing Report

Introduction

Data preprocessing is an essential step in building a deep learning model for skin disease classification. It prepares the dataset for training by improving image consistency and quality. In this project, preprocessing includes image resizing, pixel normalization, data augmentation, handling class imbalance, and dataset splitting. These steps help improve model performance, reduce overfitting, and ensure efficient training.

Dataset Overview

The **Skin Cancer MNIST (HAM10000)** dataset is used for this project. It contains dermoscopic images of seven different skin disease classes commonly used for developing and evaluating skin lesion classification models. The dataset provides labeled images that enable supervised learning for multi-class classification.

Dataset Details

Attribute	Description
Dataset Name	Skin Cancer MNIST (HAM10000)
Source	Kaggle
Total Classes	7
Image Type	Dermoscopic Images
Classification Type	Multi-Class Image Classification

Disease Classes

Class Code	Disease Name
AKIEC	Actinic Keratosis
BCC	Basal Cell Carcinoma
BKL	Benign Keratosis
DF	Dermatofibroma
MEL	Melanoma
NV	Melanocytic Nevus
VASC	Vascular Lesion

Sample Images

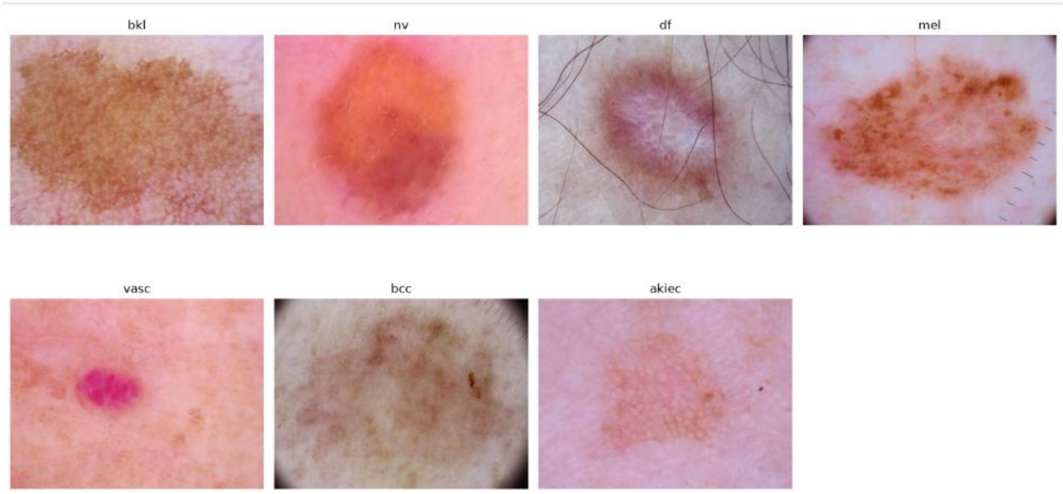


Image Resizing

To ensure a consistent input size for all deep learning models, the dermoscopic images were resized to **224 × 224 pixels** using the **OpenCV (**cv2.resize**)** function. Standardizing the image dimensions reduces computational complexity and ensures compatibility with both custom CNN and transfer learning models.

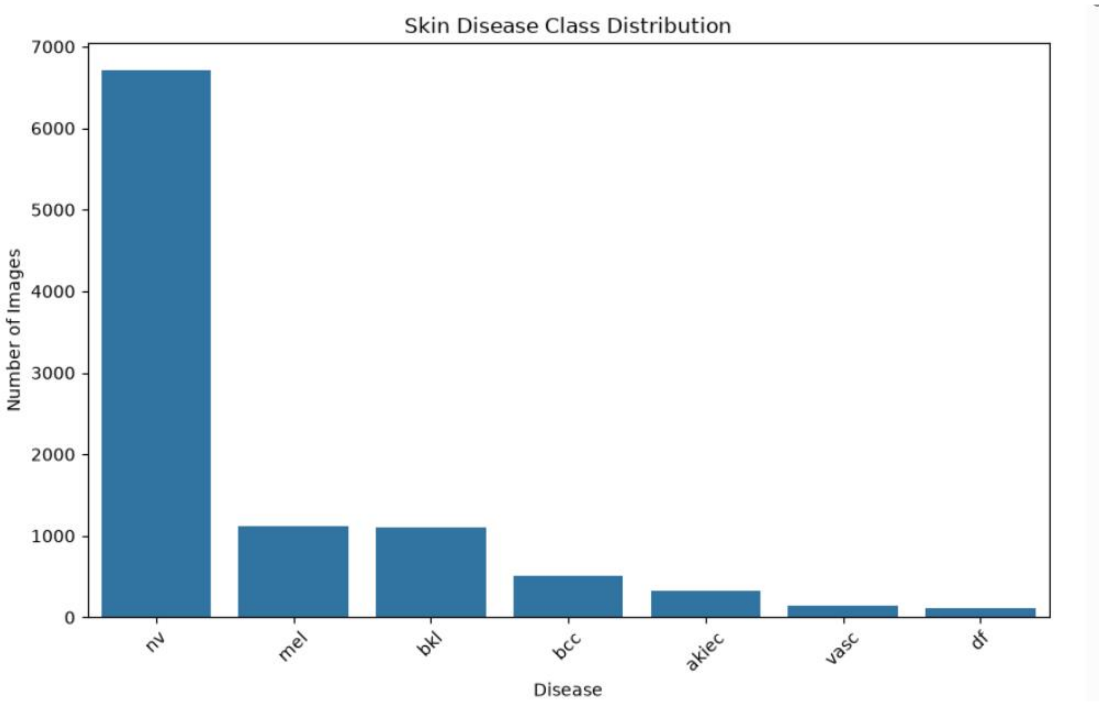
Dataset Splitting

The dataset was divided into **training**, **validation**, and **testing** sets using the train_test_split() function from Scikit-learn. A **stratified split** was applied based on the disease class (dx) to preserve the class distribution across all subsets. The random_state was set to **42** to ensure reproducibility.

Dataset Split Summary

Dataset	Shape	Percentage
Training	(7010, 8)	70%
Validation	(1502, 8)	15%
Testing	(1503, 8)	15%

Class Distribution



Handling Class Imbalance

The HAM10000 dataset contains an uneven distribution of images across the seven disease classes. To reduce prediction bias, **class weights** were computed using Scikit-learn's `compute_class_weight()` function. These weights were applied during model training to improve learning for minority classes.

Class Label Mapping

Class	Index
akiec	0
bcc	1
bkl	2
df	3
mel	4
nv	5
vasc	6

Image Augmentation

Image augmentation was applied to the training dataset using **ImageDataGenerator** to increase data diversity and improve model generalization. Validation and test datasets were only rescaled for unbiased evaluation.

Technique	Value
Rescaling	1/255
Rotation	20°
Width Shift	0.1
Height Shift	0.1
Horizontal Flip	Enabled
Vertical Flip	Enabled
Zoom	0.2
Brightness	0.8–1.2

Data Pipeline

Training, validation, and test data pipelines were created using Keras flow_from_dataframe(). Images were loaded from their file paths, resized to **224 × 224** pixels, and processed in batches of **32** images for efficient model training and evaluation.

Dataset	Target Size	Batch Size	Shuffle
Training	224 × 224	32	Yes
Validation	224 × 224	32	No
Testing	224 × 224	32	No

Result: The data pipelines efficiently loaded and preprocessed images, making them ready for deep learning model training and evaluation

Summary

The HAM10000 dataset was successfully preprocessed by resizing images to **224 × 224** pixels, normalizing pixel values, applying image augmentation, handling class imbalance using class weights, and splitting the data into training, validation, and testing sets. The processed data pipelines were prepared for efficient and reliable deep learning model training.